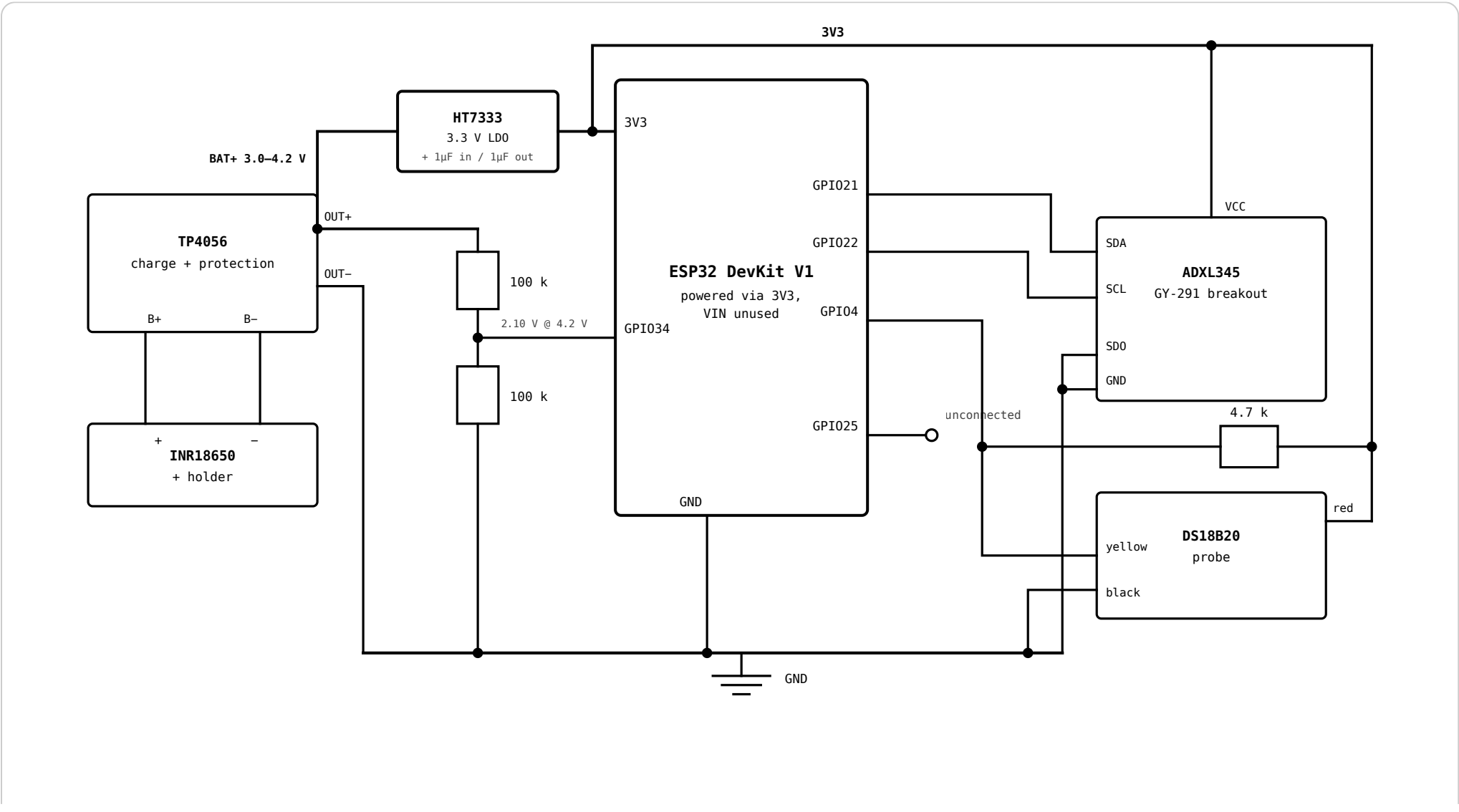
Prototype wiring — ESP32 DevKit V1 + ADXL345 + INR18650

The bench rig, drawn at module level. This is *not* the custom PCB schematic (`schematic-gravity-module.html`) — that one shows bare SMD parts you can't breadboard. Every line below is a wire you physically run.



Every line is a wire. A filled dot is a junction; lines crossing without a dot are not connected (the 4.7 kΩ pull-up crosses the ADXL345's ground drop at one point). **SDO is tied to GND** — that is what puts the ADXL345 at address `0x53`, which is the address the firmware expects.

It must be an ESP32 board. "D1 Mini" is sold in both an ESP8266 and an ESP32 version, which look nearly identical and are often listed under the same name. The firmware uses `<WiFi.h>`, `RTC_DATA_ATTR`, `esp_sleep_enable_timer_wakeup()` and a 12-bit 3.3 V ADC — all ESP32-only, so on an ESP8266 it does not compile and the 100k/100k divider would also over-range that board's 1.0 V ADC input. Check the module marking, or what it flashes as in the Arduino IDE.

GPIO numbers, not silkscreen labels. Board revisions and clones print these pins differently (`IO4`, `D4`, plain `4`). The firmware addresses GPIO numbers; match those. Both the D1 Mini ESP32 and the DevKit V1 break out all five.

On a DevKit V1 this power chain does not work. Its AMS1117 regulator needs >4.4 V in to hold 3.3 V out, and a cell is 4.2 V falling — you get brownouts during WiFi transmit that look like random resets. Either regulate the cell to 3.3 V with an HT7333/MCP1700 and feed `3V3` directly, or use the D1 Mini, whose regulator runs from a cell as drawn. **Never feed the raw cell into `3V3`** (4.2 V against a 3.6 V maximum), and never have USB connected at the same time as an external 3V3 feed.

Connections

From	To	Why
18650 + / −	TP4056 `B+` / `B−`	Cell into the charger
TP4056 `OUT+`	HT7333 input	The DevKit V1's own AMS1117 can't hold 3.3 V from a cell — this LDO replaces it
HT7333 output	Board `3V3` + sensor rail	Regulated 3.3 V. `VIN` / `5V` stays unused. **Never connect USB at the same time**
TP4056 `OUT+`	100 k → GPIO34 → 100 k → GND	Battery sense. 2.10 V at a 4.2 V cell, matching the firmware's ×2
TP4056 `OUT−`	GND bus	Common ground
GPIO21 / GPIO22	ADXL345 `SDA` / `SCL`	I²C

ADXL345 `VCC`	Board `3V3`	
ADXL345 `SDO`	GND	**Sets address 0x53.** Floating or high gives 0x1D and the sketch reports no samples
GPIO4	DS18B20 yellow	1-Wire data
GPIO4	4.7 kΩ → `3V3`	Bus pull-up. Some probe modules already have one — don't fit two
DS18B20 red / black	`3V3` / GND	
GPIO25	— nothing	Divider gating isn't fitted on a breadboard; the firmware toggles it harmlessly

None of this is needed for the axis test. Board + ADXL345 + USB cable is enough to confirm the sketch prints ~0° upright and ~90° flat — the assumption the whole firmware rests on. Wire the battery and DS18B20 afterwards.